Unit - II

Process management

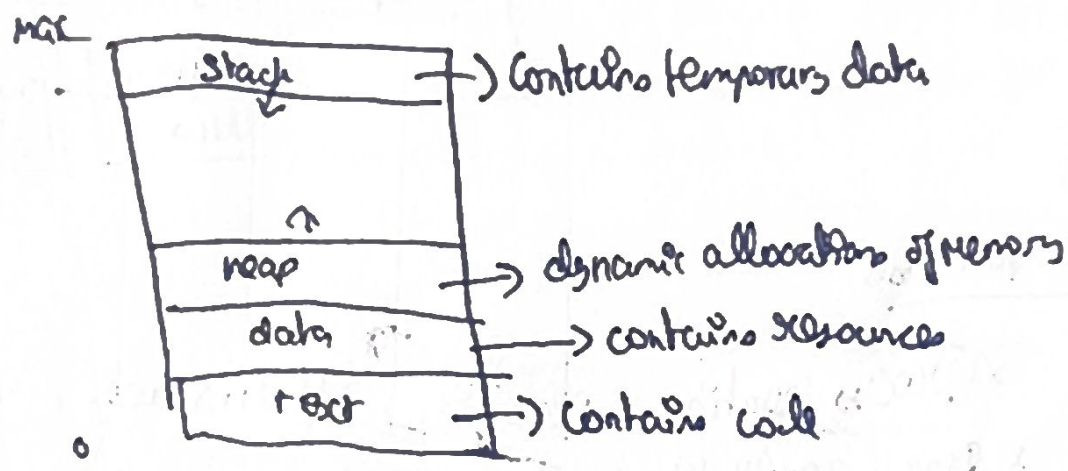
Topics covered:

*) Process concept ✓	*) Semaphores ×
*) Process States ✓	*) Synchronization issues] ×
*) Operations on processes	*) Deadlock system model
*) CPU Scheduling ✓	*) Deadlock prevention ×
*) Scheduling criteria ✓	*) Detection and recovery
*) Scheduling algorithm ✓	*) User and group management concepts
*) Process synchronization	*) Overviews of system monitoring and performance utilities ×
*) Critical Section problems	

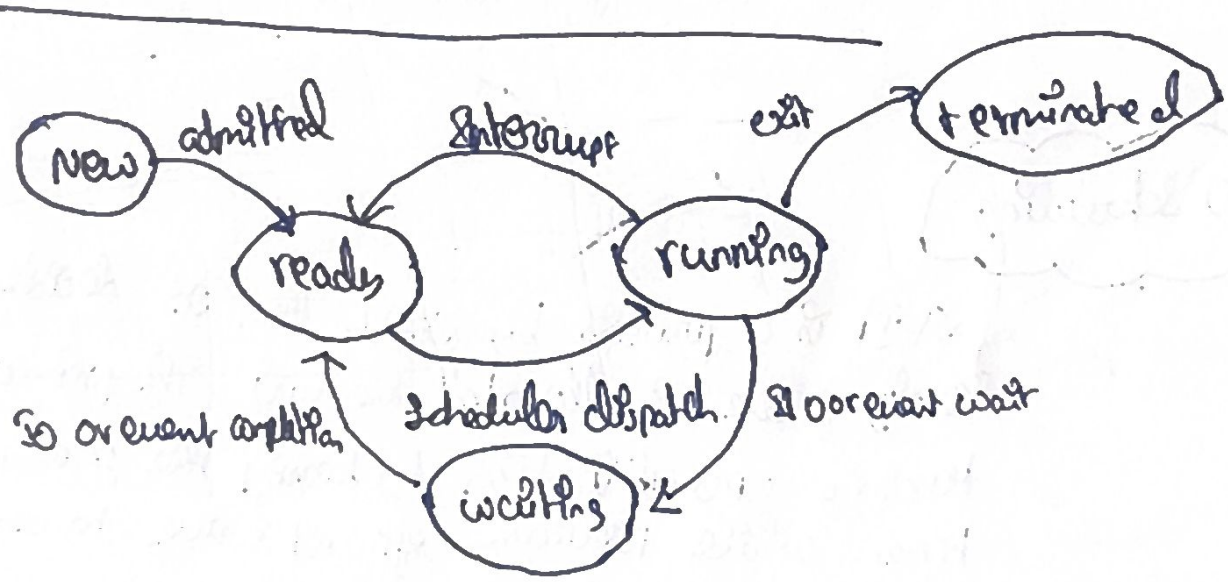
Process

*) It is the performance of the program / execution

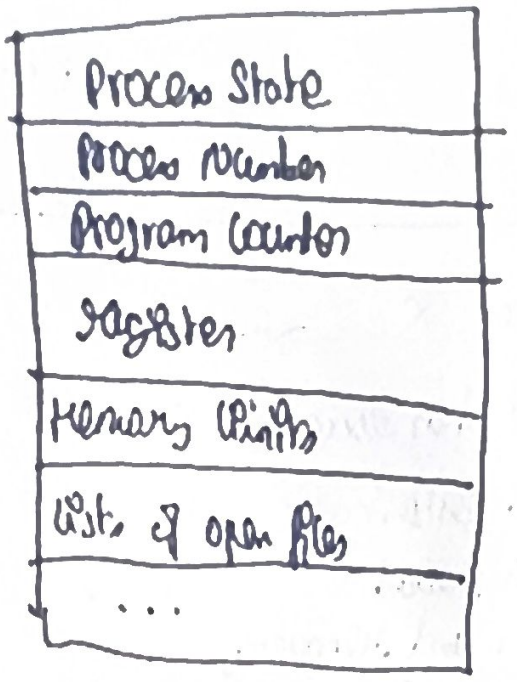
A process in memory is organized as



As process executes it move through different states



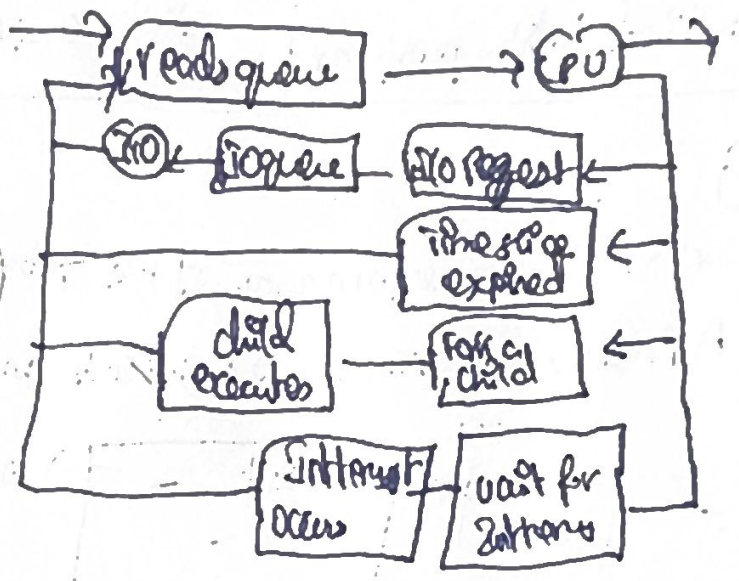
Process control Block



Process Scheduling :

* Long term scheduler
* Short term scheduler

* I/O bound process
* CPU bound process



Operations on processes

* Process creation (execution, address space, system calls)
* Process Termination (exit(), wait())

CPU Scheduling

* It is a process by which the OS decides which process in the ready queue is allocated the CPU. The primary goal is to maximize CPU utilization by keeping the processor busy at the time while providing efficient service to users

The CPU - I/O Burst cycle

Process execution follows a cycle where a process alternates between CPU bursts and I/O bursts. A process begins with a CPU burst, followed by an I/O burst, and continues this pattern until its final CPU burst ends with a request to terminate

Preemptive vs non preemptive scheduling

*) Scheduling decisions occur when a process switches occur

*) Non preemptive scheduling: A process keeps the CPU until it either terminates or requests I/O to enter a waiting state

*) preemptive scheduling: The OS can interrupt a running process and execution to take the CPU away and allocate it to another process

Scheduling criteria:

To evaluate the effectiveness of different algorithm

- CPU utilization
- Throughput: The number of process completed per unit of time
- Turnaround time: The total time from process submission to completion (waiting time + burst time)
- waiting time: The sum of all periods a process spends waiting in the ready queue
- Response time: The time from submission of a request until the first response is produced

CPU Scheduling Algorithms:

1) First come, First served (FCFS) : Disadvantage: convoy effect long waiting time
2) Shortest Job First (SJF) : Starvation (long process)
3) priority scheduling: low priority ⇒ Starvation, solution Aging
4) Round Robin : all tasks are atleast executed once.

Process synchronization

* It deals with the coordination and execution of multiple processes in such a way that they do not interfere with each other while sharing resources.
* It ensures the consistency of shared data and prevents data race, deadlock and other synchronization

Need: when multiple processes execute concurrently, they may need to access shared resources like memory file or I/O devices

* If these accesses are not controlled, problems such as

i) Race Conditions : multiple process access & manipulation at the same data concurrently
final outcome depends on their time of execution

ii) Data Inconsistency: Shared data may become inconsistent due to uncontrol access

iii) Deadlock : Process waits indefinitely waiting for resources held by each other

iv) Starvation : A process waits indefinitely because other process are continuously given preference.

Critical Section problem :

* The critical section is a part of a program where shared resources are accessed
* It ensures that only one process enters the critical section at a time.

Solution Requirements :

i) Mutual exclusion: only one process should be allowed in the critical section at any time

ii) Progress : If no process is in the critical section and some process wish to enter, one of them should be allowed to proceed.